

Sagar Hasan

Software Engineer | React + TypeScript Specialist

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[GitHub](#) | [LinkedIn](#) | [LeetCode](#) | [Portfolio](#)

SUMMARY

Software Engineer with **3 years** of frontend-focused experience building scalable React/TypeScript applications. **Competitive programmer with 1,200+ problems solved on LeetCode** – demonstrating strong command of data structures, algorithms, and performance optimization. Experienced in full development lifecycle including API integration, system design collaboration, and cross-functional agile teams. Passionate about crafting high-performance, responsive UIs, **mentoring new developers**, conducting code reviews, and writing clean, maintainable code

TECHNICAL SKILLS

- **Frontend Core:** React.js, Next.js, TypeScript, JavaScript (ES6+), JSX, Babel
- **State Management:** Redux, Redux Toolkit, Context API
- **UI & Styling:** Material-UI, HTML5, CSS3, Flexbox/Grid, Responsive Design
- **Frontend Tooling:** Webpack, Babel, npm/yarn, Mocha, Chrome DevTools
- **API & Real-Time:** RESTful APIs, Socket.io, Axios, React Query
- **Backend Awareness:** Node.js, Express.js (for API debugging & collaboration)
- **Database Knowledge:** MySQL, MongoDB (for understanding data flow)
- **Additional:** Data Structures & Algorithms, DOM Manipulation, Virtual DOM, Component Lifecycle, Performance Profiling

EXPERIENCE

Software Engineer | LYXA S.A.L (Gulshan 1, Dhaka)

07/2024 – present

- Designed and developed **reusable React components** with TypeScript and Material-UI, ensuring consistency across 10+ product modules.
- Managed complex application state using **Redux Toolkit**, reducing prop-drilling and improving component reusability by 30%.
- Integrated **RESTful APIs** and Socket.io for real-time data synchronization, delivering a seamless live-updating user experience.
- Profiled frontend performance using Chrome DevTools; optimized component re-renders using **React.memo**, **useCallback**, and **useMemo**, improving page load speed by 25%.
- Mentored 3 junior developers through weekly 1-on-1 sessions and pair programming – helping them ship features independently within 2 months.

Software Engineer Intern | LYXA S.A.L

01/2024 – 06/2024

- Debugged UI bugs and cross-browser compatibility issues, ensuring consistent rendering across Chrome, Firefox, and Safari.
- Built and maintained interactive frontend features using **React and TypeScript**.
- Participated in daily stand-ups and sprint planning, **delivering 5+ frontend** features on schedule.

Software Engineer Intern (Project-Based) | DIU (Ashulia, Savar, Dhaka)

Algorithms Visualizer – Internship Project | 07/2023 - 12/2023

- Designed and developed **Algorithms Visualizer**, a React-based educational platform
- Built **15+ algorithm and data structures** visualizations (array, stack, sorting, pathfinding, tree traversals) using Canvas API and SVG, translating complex DSA concepts into interactive visual experiences.
- Implemented animation controls (speed, pause, step-by-step) using React state and hooks, improving student comprehension by 40% (based on feedback).

PROJECTS (System Design & Real-Time Focus)

Val5th-Finder - Full Stack

[Client-Repo](#) [Server-Repo](#) [Live](#)

Developed a full-stack real-time lobbymaking platform for Valorant players to find teammates ("5th player") based on rank, roles, and playstyle.

- **Frontend:** React 18, TypeScript, Redux Toolkit, Material-UI (MUI), Framer Motion, React Hook Form, Zod, Axios, Socket.io-client, Vite
- **Backend:** Node.js, Express, MongoDB, TypeScript, Socket.io, JWT Authentication
- Designed a **Lobby distribution algorithm** to match players based on rank, roles, region, and availability.
- Designed as two loosely coupled services (lobby service + match service).

Talk2Active (Language practice platform)

[Client-Repo](#) [Server-Repo](#) [Live](#)

- **Frontend:** React.js, Redux, Material-UI, TypeScript, Zod, Axios, Socket.io-client, WebRTC
- **Backend:** Node.js, Express, MongoDB, TypeScript, Socket.io, WebRTC, JWT Authentication
- Engineered **Post distribution algorithm** and **Room-splitting algorithm** to optimize task allocation and improve system scalability.

File Processing kit - No Data Uploads Required

[Git Repo](#) [Live](#)

- React hooks, Canvas API, pdf-lib, Tesseract.js – shows strong frontend + utility building.

FRONTEND QUALITY & OBSERVABILITY

- Wrote unit tests for React components using Mocha, achieving 85% test coverage.
- Use **Chrome DevTools** and **Lighthouse** to audit and improve frontend performance and accessibility.
- Report UI bugs with clear reproduction steps and video recordings.
- Documented component library using Storybook and JSDoc, improving team onboarding efficiency by 30%.

EDUCATION

Bachelor of Science: Computer Science and Engineering

Daffodil International University - Savar, Dhaka, Bangladesh

September, 2019 - February, 2024

Relevant coursework: Data Structures, Algorithms, OOP, Databases (MySQL), OS, Networking

Languages: English (Professional Working Proficiency), Bengali (Native).

Interests: Open-source contributions, UI/UX design, mentoring, and gaming.